OFFICIAL ARCHITECTURE SPECIFICATION & KYC GUIDE

PhonePe Payment Gateway
Integration Blueprint

Comprehensive Architecture, Merchant KYC, Security Protocol & Production Setup for **g1mart.in**

APPLICATION	PRODUCTION DOMAIN
G1 Mart Supermarket Web App	**https://g1mart.in**
PAYMENT GATEWAY	TECH STACK
PhonePe Standard Web Checkout (PG v1)	**Next.js 15.3, React 19, Supabase**
MERCHANT LOCATION	BLUEPRINT VERSION
Padarupalli, Nellore, AP - 524004	**v1.0 (Production Master) — Oct 2026**

EXECUTIVE SUMMARY & WHY PHONEPE

PhonePe is India's leading digital payments platform with over 48% market share in UPI. Integrating PhonePe Standard Checkout for **g1mart.in** provides G1 Mart customers with instant, zero-friction UPI app intent payments (PhonePe, Google Pay, Paytm, BHIM), RuPay Debit, Credit Cards, and NetBanking.

The G1 Mart codebase already contains an architecture-grade PhonePe driver at src/lib/payment/phonepeAdapter.ts with built-in HMAC SHA-256 (X-VERIFY) security, automatic paise conversion, and dual-channel verification (Server-to-Server Webhooks + Browser Order Status Polling).

KEY DELIVERABLES IN THIS BLUEPRINT

- **Section 1: Merchant KYC & Compliance Checklist** — Documents needed to obtain live PhonePe credentials.
- **Section 2: Mandatory Website Compliance on g1mart.in** — Required policy pages (Refunds, Privacy, Terms).
- **Section 3: Production Endpoint Directory** — Complete routing table for APIs, webhooks, and return URLs.
- **Section 4: Technical Lifecycle Architecture** — 7-step sequence diagram from cart to live delivery dispatch.
- **Section 5: Cryptographic Security (X-VERIFY)** — Formula and code for SHA-256 checksum calculation.
- **Section 6: Code Walkthrough & Files to Update** — Exact file modifications to link the storefront to PhonePe.
- **Section 7: Step-by-Step Testing & Go-Live Checklist** — Sandbox testing to live production settlement.

1. Merchant Account & Legal KYC Requirements

To obtain Production Live credentials (Merchant ID & Salt Key), the store owner must sign up at **https://business.phonepe.com** and submit business documents for compliance verification.

Document	Requirement Details	Purpose / Verification
GST Certificate	GSTIN Registration certificate for G1 Mart Supermarket (Nellore, AP)	Legal entity verification
Business PAN	Company PAN card (or Proprietor's PAN card for Sole Proprietorship)	RBI tax compliance & banking identity
Bank Account Proof	Cancelled Cheque or latest 3-month Bank Statement showing Account Name, Number & IFSC	Daily automated payout settlement account (T+1 cycle)
Signatory KYC	Aadhaar Card, Passport, or Voter ID of authorized store owner	Identity & address verification of owner
Storefront Photos	Exterior signboard photo ('G1 Mart Supermarket'), interior aisles, and Padarupalli location	Physical merchant audit verification

2. Website Compliance Checklist on https://g1mart.in

PhonePe's risk team verifies the live website before turning on live transactions. The following information must be publicly accessible on **https://g1mart.in**:

Policy / Feature	Required Content on g1mart.in	Current Codebase Status
Contact Us & Store Info	Physical store address (Padarupalli, Nellore, AP - 524004), Phone (+91 91107 59629), Email (support@g1mart.in)	✓ **Implemented** (store.ts & Footer)
Pricing in INR (■)	All products display clear selling prices in Indian Rupees without hidden charges	✓ **Implemented** (Catalog & Cart)
Shipping & Delivery	Estimated delivery time (15-30 mins in Nellore), serviceable 30km radius, minimum order details	✓ **Implemented** (Checkout zone engine)
Terms & Conditions	Customer user agreement, order acceptance, Nellore jurisdiction clause (Page: /terms)	■ **Publish static page**
Privacy Policy	User data protection, phone/address usage, statement that payment credentials are encrypted by PhonePe (Page: /privacy)	■ **Publish static page**
Refund & Cancellation	Cancellation permitted before order packed; refund timeline 5-7 business days via original UPI mode (Page: /refund-policy)	■ **Publish static page**
SSL Security	Active HTTPS on https://g1mart.in with 256-bit encryption	✓ **Standard on Hosting**

3. Production URLs & Endpoint Directory (g1mart.in)

Below is the exact production URL mapping between customer devices, the G1 Mart Next.js application, and PhonePe servers:

Endpoint	Full Production URL	HTTP	Role & Action
Storefront Root	https://g1mart.in	GET	Base website URL configured in NEXT_PUBLIC_APP_URL
Checkout Page	https://g1mart.in/checkout	GET	Address selection, zone minimum order check
Payment Page	https://g1mart.in/payment	GET	Payment selection (Pay Online vs Pay on Delivery)
Initiate API	https://g1mart.in/api/checkout/initiate	POST	Calculates cart from DB, calls PhonePe /pg/v1/pay
Customer Return	https://g1mart.in/payment/result?orderId=...	GET/POST	Redirect destination after customer completes payment on PhonePe
S2S Webhook	https://g1mart.in/api/payment/webhook	POST	Server-to-server callback registered on PhonePe Merchant Portal
Status Poll API	https://g1mart.in/api/orders/{id}/payment-status	GET	Polls PhonePe Order Status API (/pg/v1/status/{mid}/{id})
Order Tracking	https://g1mart.in/orders/{id}?paid=true	GET	Post-checkout invoice, live packing status & rider GPS track

Required Environment Variables (.env.local & Production Hosting)

```
# Base Application URL NEXT_PUBLIC_APP_URL=https://g1mart.in # PhonePe Production Credentials
PHONEPE_MERCHANT_ID=G1MARTOFFLINE PHONEPE_SALT_KEY=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx PHONEPE_SALT_INDEX=1
PHONEPE_ENV=PRODUCTION # Optional Webhook Basic Auth (if set in PhonePe Portal)
PHONEPE_WEBHOOK_USERNAME=g1mart_s2s_user PHONEPE_WEBHOOK_PASSWORD=g1mart_s2s_password
```

4. Technical Payment Lifecycle & Architecture

PhonePe integration uses a **Dual-Channel Settlement Pattern** ensuring 100% order capture even if the customer's phone disconnects:

Step 1: Initiation: Customer selects 'Pay online' and clicks checkout. The browser calls POST /api/checkout/initiate with cart items and address.

Step 2: Server Security Recalculation: serverOrderStore recalculates item subtotal from the database. It rejects any client price manipulations. Order ID (G1-XXXXXX) and Merchant Transaction ID (MT_G1-XXXXXX_time) are assigned.

Step 3: PhonePe Pay Page API: PhonePePaymentAdapter converts payload to Base64, computes SHA-256 X-VERIFY checksum with the Salt Key, and posts to PhonePe /pg/v1/pay. PhonePe returns a secure payment redirect URL.

Step 4: Browser Redirect to PhonePe: The customer is seamlessly redirected to PhonePe's hosted pay page. Customer selects UPI Intent (PhonePe/GPay), UPI QR, RuPay Card, or NetBanking.

Step 5: Channel A — Server-to-Server Webhook: PhonePe immediately fires an asynchronous POST to https://g1mart.in/api/payment/webhook. The server validates the checksum signature, checks idempotency, and updates Supabase orders.payment_status = 'completed'.

Step 6: Channel B — Return Redirect & Status Poll: Customer browser returns to https://g1mart.in/payment/result?orderId=G1-XXXXXX. If webhook has completed, page confirms instantly. If pending, it polls /api/orders/{id}/payment-status every 2.5s.

Step 7: Real-Time Store Fulfillment: Confirmed order broadcasts to G1 Mart store tablet and packing screen for 15-30 minute dispatch.

5. Cryptographic Security (X-VERIFY Checksum)

PhonePe requires SHA-256 HMAC checksums on every request and webhook callback to prevent tampering:

1. Pay Page Initiation Header (POST /pg/v1/pay)
X-VERIFY: SHA256(base64Payload + "/pg/v1/pay" + PHONEPE_SALT_KEY) + "###" + PHONEPE_SALT_INDEX

2. Order Status Polling Header (GET /pg/v1/status/{mid}/{transactionId})
X-VERIFY: SHA256("/pg/v1/status/" + PHONEPE_MERCHANT_ID + "/" + transactionId + PHONEPE_SALT_KEY) + "###" + PHONEPE_SALT_INDEX

3. S2S Webhook Signature Verification
calculated = SHA256(req.body.response + PHONEPE_SALT_KEY) + "###" + PHONEPE_SALT_INDEX
If received_x_verify !== calculated, the webhook is rejected with HTTP 400.

6. Code Walkthrough & Files to Connect

The G1 Mart codebase already has the complete PhonePe adapter built. To activate PhonePe end-to-end, update these three existing files:

1. **src/app/api/checkout/initiate/route.ts**: Switch the payment adapter call from razorpayAdapter to phonePeAdapter. PhonePe returns redirectUrl which is passed back to the client.
2. **src/app/(storefront)/payment/page.tsx**: When user clicks 'Pay Online' and API responds with { success: true, redirectUrl }, simply redirect the browser window: window.location.href = data.redirectUrl.
3. **src/app/(storefront)/payment/result/page.tsx**: Page already reads ?orderId=G1-XXXXXX and polls /api/orders/[id]/payment-status. When phonePeAdapter confirms payment, it transitions to confirmed state.

7. Testing & Go-Live Activation Checklist

Step	Action Item	Verification Check
1. Sandbox Test	Set PHONEPE_MERCHANT_ID=PGTESTPAYUAT, PHONEPE_ENV=SANDBOX in .env.local	API hits PhonePe UAT preprod endpoint; test transactions succeed
2. Merchant Portal	Sign up on business.phonepe.com, submit GSTIN, Bank Proof & Aadhaar	KYC verification approved
3. Production Keys	Generate Production Merchant ID & 32-character Salt Key in Developer Tab	Save securely in production environment
4. Register Webhook	Enter Webhook URL: https://g1mart.in/api/payment/webhook	PhonePe sends test ping with HTTP 200 response
5. Whitelist Domain	Add https://g1mart.in under Allowed Domains in PhonePe Portal	Domain status displays 'Approved'
6. Live Small Order	Perform real transaction of ■10 (e.g. soap or salt pack) using PhonePe UPI app	Order marked Paid, status updated to Order Placed, webhook returns HTTP 200
7. Bank Payout	Verify daily settlement report on PhonePe Merchant Dashboard	Net amount credited to G1 Mart bank account (T+1 day)

Official PhonePe Support Contacts
- **Merchant Support Email**: merchant-support@phonepe.com
- **Technical Integration Support**: pgsupport@phonepe.com
- **Developer Docs**: https://developer.phonepe.com/v1/docs/standard-checkout-web